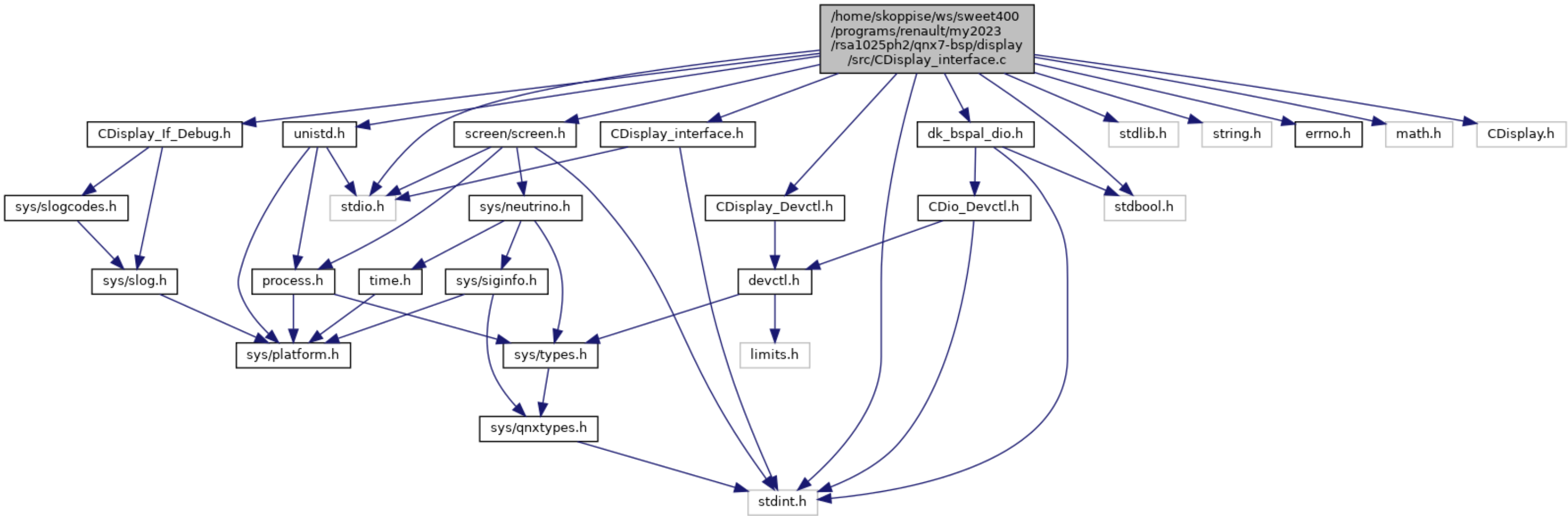


CDisplay_interface.c File Reference

```
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <string.h>
#include <stdbool.h>
#include <errno.h>
#include <unistd.h>
#include <screen/screen.h>
#include <math.h>
#include "CDisplay_interface.h"
#include "CDisplay_If_Debug.h"
#include "dk_bspal_dio.h"
#include "CDisplay_Devctl.h"
#include "CDisplay.h"
```

Include dependency graph for CDisplay_interface.c:



Macros

```
#define GAMMA_NUM_STEPS 256U
#define GAMMA_MAX_VALUE 65535U
#define SCALING_FACTOR 100.0f
```

Functions

static int32_t	Init_Display (void) Display Gpios configuration. More...
static int32_t	Enable_Display (uint8_t *p_enable_display_U8) To enable and disable the Display Gpios configuration. More...
static int32_t	Get_Display_Power (uint8_t *p_power_U8) To get the display power value. More...
static int32_t	Get_Display_Power_Devctl (uint8_t *p_value_U8) To get the display power value. More...
static int32_t	Get_Display_state (uint8_t *p_state_U8) To get the display state. More...
static int32_t	Get_Display_state_Devctl (uint8_t *p_value_U8) To get the display state. More...
static int32_t	Get_Display_Thermal_Temperature_Value (uint8_t *p_value_U8) To get the display Thermal temperature value. More...
static int32_t	Get_Display_Thermal_Temperature_Value_Devctl (uint32_t *p_temp_U32) To get the display Thermal temperature value. More...
static int32_t	Deinit_Display (void) To deinit display. More...
static int32_t	Set_Display_State (uint8_t p_onflag) control the display power and LVDS state, More...
static int32_t	Set_Gamma_RGB_Devctl (int32_t p_gamma_red_S32, int32_t p_gamma_green_S32, int32_t p_gamma_blue_S32) control the display Gamma Factor for RED, GREEN, BLUE, More...
static int32_t	Get_Gamma_RGB_Devctl (int32_t *p_gamma_red_S32, int32_t *p_gamma_green_S32, int32_t *p_gamma_blue_S32)
void	generate_gamma_table (double gamma, int32_t resolution, int32_t size, int32_t *arr) Generate the Gamma Array of 256 for the gamma factor input. More...

Variables

static bool	lcd_on_BOOL = false
double	g_gamma_red_u32 =0
double	g_gamma_green_u32 =0
double	g_gamma_blue_u32 =0
ts_display_funcs	display_drv_entry

Macro Definition Documentation

◆ GAMMA_MAX_VALUE

#define GAMMA_MAX_VALUE 65535U

◆ GAMMA_NUM_STEPS

```
#define GAMMA_NUM_STEPS 256U
```

◆ SCALING_FACTOR

```
#define SCALING_FACTOR 100.0f
```

Function Documentation

◆ Deinit_Display()

```
static int32_t Deinit_Display ( void )
```

static

To deinit display.

Function Name: Deinit_Display

Parameters

[in] **none**

Returns

int32_t Error code to indicate success or failure

◆ Enable_Display()

```
static int32_t Enable_Display ( uint8_t * p_enable_display_U8 )
```

static

To enable and disable the Display Gpios configuration.

Function Name: Enable_Display

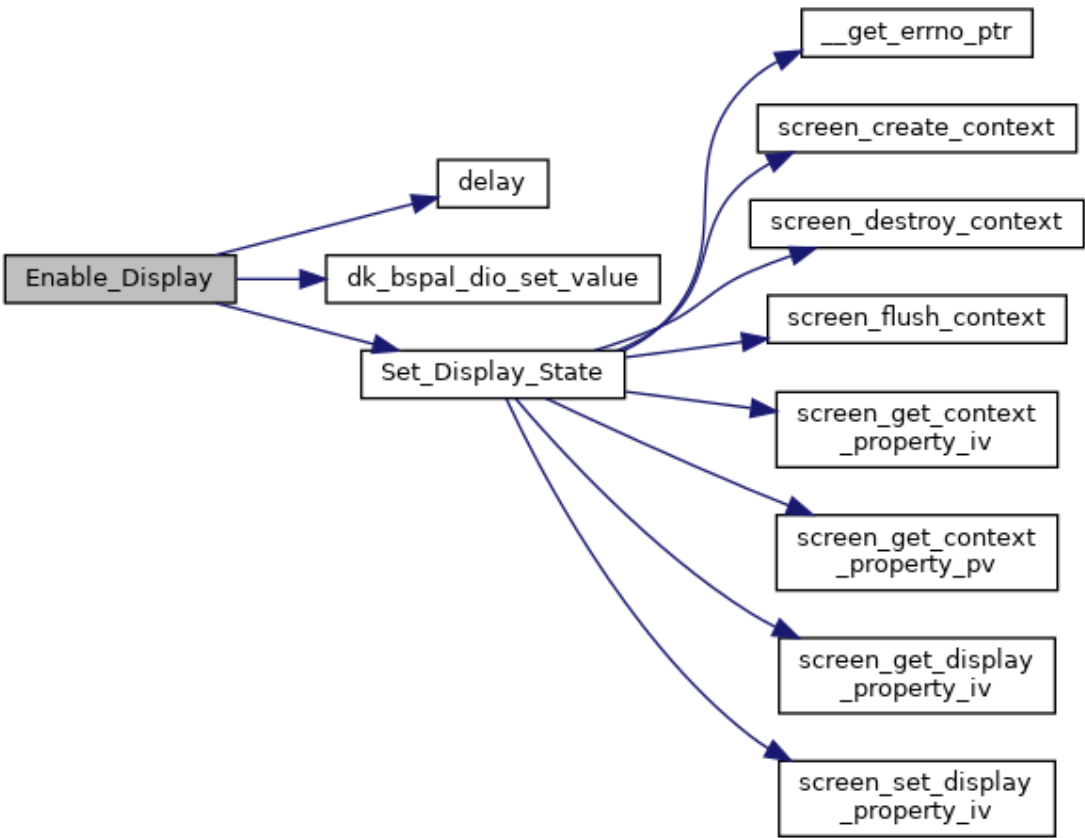
Parameters

[in] *p_enable_display_BOOL - bool indicates whether to enable or disable the display

Returns

int32_t Error code to indicate success or failure

Here is the call graph for this function:



◆ generate_gamma_table()

```
void generate_gamma_table ( double  gamma,
                           int32_t  resolution,
                           int32_t  size,
                           int32_t * arr
                           )
```

Generate the Gamma Array of 256 for the gamma factor input.

Function Name: generate_gamma_table

Parameters

[in] **gamma** - Gamma Factor resolution- Max Value for the Gamma Curve size - Gamma Curve Steps

Returns

Null

Here is the caller graph for this function:



◆ Get_Display_Power()

```
static int32_t Get_Display_Power ( uint8_t * p_power_U8 )
```

static

To get the display power value.

Function Name: Get_Display_Power

Parameters

[in] ***p_value_power_U8** - uint8_t power value to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Display_Power_Devctl()

static int32_t Get_Display_Power_Devctl (uint8_t * p_value_U8)

static

To get the display power value.

Function Name: Get_Display_Power_Devctl

Parameters

[in] *p_value_power_U8 - uint8_t power value to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Display_state()

static int32_t Get_Display_state (uint8_t * p_state_U8)

static

To get the display state.

Function Name: Get_Display_state

Parameters

[in] *p_state_U8 - uint8_t power state to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Display_state_Devctl()

static int32_t Get_Display_state_Devctl (uint8_t * p_value_U8)

static

To get the display state.

Function Name: Get_Display_state_Devctl

Parameters

[in] *p_state_U8 - uint8_t power state to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Display_Thermal_Temperature_Value()


```
static int32_t Get_Display_Thermal_Temperature_Value ( uint8_t * p_value_U8 )
```

static

To get the display Thermal temperature value.

Function Name: Get_Display_Thermal_Temperature_Value

Parameters

[in] ***p_state_U8** -uint8_t temperature value to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Display_Thermal_Temperature_Value_Devctl()

```
static int32_t Get_Display_Thermal_Temperature_Value_Devctl ( uint32_t * p_temp_U32 )
```

static

To get the display Thermal temperature value.

Function Name: Get_Display_Thermal_Temperature_Value_Devctl

Parameters

[in] ***p_state_U8** -uint8_t temperature value to get

Returns

int32_t Error code to indicate success or failure

◆ Get_Gamma_RGB_Devctl()

```
static int32_t Get_Gamma_RGB_Devctl ( int32_t * p_gamma_red_S32,
                                     int32_t * p_gamma_green_S32,
                                     int32_t * p_gamma_blue_S32
                                     )
```

static

◆ Init_Display()

static int32_t Init_Display (void)

static

Display Gpios configuration.

Function Name: Init_Display

Parameters

[in] **void**

Returns

int32_t Error code to indicate success or failure

Here is the call graph for this function:



◆ Set_Display_State()


```
static int32_t Set_Display_State ( uint8_t p_onflag )
```

static

control the display power and LVDS state,

Function Name: Set_Display_State

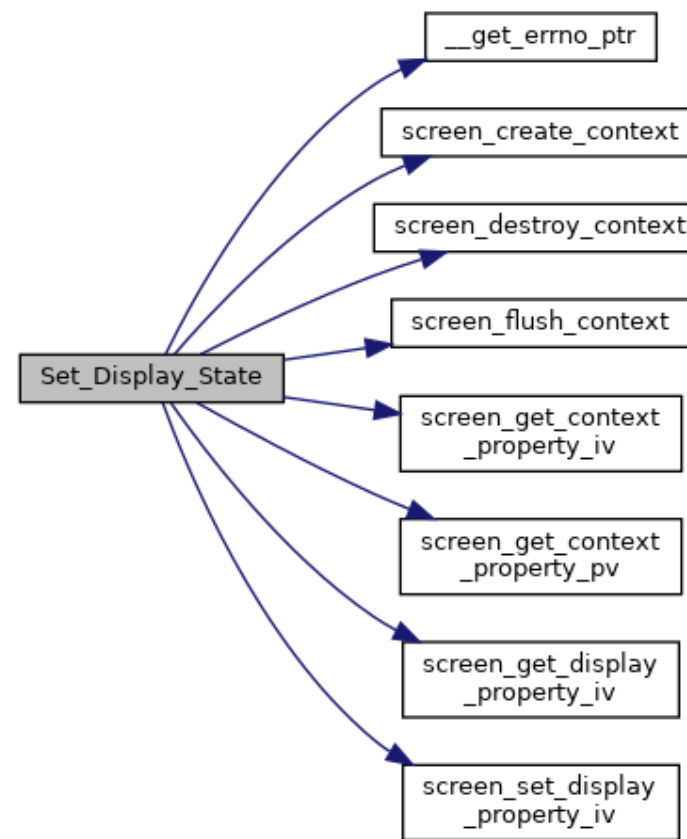
Parameters

[in] **p_onflag** - set to 1 to turn on, 0 for off

Returns

int32_t Error code to indicate success or failure

Here is the call graph for this function:



Here is the caller graph for this function:



◆ Set_Gamma_RGB_Devctl()

```
static int32_t Set_Gamma_RGB_Devctl ( int32_t p_gamma_red_S32,  
                                     int32_t p_gamma_green_S32,  
                                     int32_t p_gamma_blue_S32  
                                     )
```

static

control the display Gamma Factor for RED, GREEN, BLUE,

Function Name:Set_Gamma_RGB_Devctl

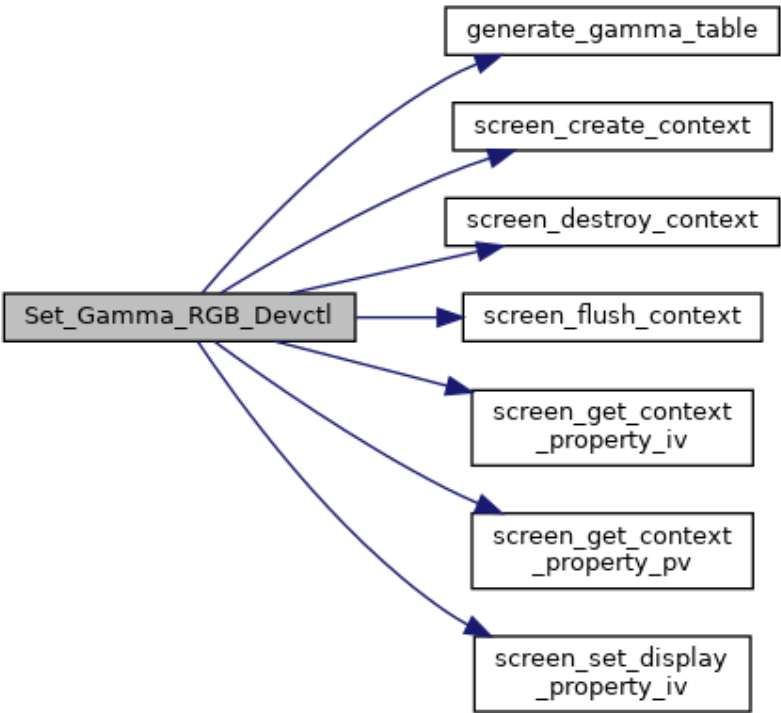
Parameters

[in] **p_gamma_red_S32,p_gamma_green_S32,p_gamma_blue_S32** - Set the Display Property - Gamma Correction respectively for RGB

Returns

int32_t Error code to indicate success or failure

Here is the call graph for this function:



Variable Documentation

◆ display_drv_entry

Initial value:

```
=
{
    sizeof ( ts_display_funcs ),
    Init_Display,
    Enable_Display,
    Get_Display_Power,
    Get_Display_Power_Devctl,
    Get_Display_state,
    Get_Display_state_Devctl,
    Get_Display_Thermal_Temperature_Value,
    Get_Display_Thermal_Temperature_Value_Devctl,
    Deinit_Display,
    Set_Gamma_RGB_Devctl,
    Get_Gamma_RGB_Devctl
}
```

◆ g_gamma_blue_u32

```
double g_gamma_blue_u32 =0
```

◆ g_gamma_green_u32

```
double g_gamma_green_u32 =0
```

◆ g_gamma_red_u32

```
double g_gamma_red_u32 =0
```

◆ lcd_on_BOOL

```
bool lcd_on_BOOL = false
```

static